Supplementary Material:

A systematic review and network meta-analysis of interventions for subjective cognitive decline

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Table 1: The PRISMA for Abstracts Checklist

TITLE	**CHECKLIST ITEM**	REPORTED ON PAGE #
1. Title:	Identify the report as a systematic review, meta-analysis, or both.	1
BACKGROUND		
2. Objectives:	The research question including components such as participants, interventions, comparators, and outcomes.	2
METHODS		
3. Eligibility criteria:	Study and report characteristics used as criteria for inclusion.	2
4. Information sources:	Key databases searched and search dates.	2
5. Risk of bias:	Methods of assessing risk of bias.	2
RESULTS		
6. Included studies:	Number and type of included studies and participants and relevant characteristics of studies.	2
7. Synthesis of results:	Results for main outcomes (benefits and harms), preferably indicating the number of studies and participants for each. If meta-analysis was done, include summary measures and confidence intervals.	2
8. Description of the effect:	Direction of the effect (i.e. which group is favoured) and size of the effect in terms meaningful to clinicians and patients.	2
DISCUSSION		
9. Strengths and limitations of evidence:	Brief summary of strengths and limitations of evidence (e.g. inconsistency, imprecision, indirectness, or risk of bias, other supporting or conflicting evidence)	2
10. Interpretation:	General interpretation of the results and important implications	2
OTHER		
11. Funding:	Primary source of funding for the review.	1
12. Registration:	Registration number and registry name.	2

Table 2: The PRISMA checklist for systematic reviews

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Suppl. Material
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5 & 6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8 & Suppl. Material

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	11
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	11 & 12
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9 & Suppl. Material
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	11
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9 - 11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Suppl. Material
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Suppl. Material
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12-14
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15 &16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	17

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

Table 3: Interventions for people with subjective cognitive decline, search string in **MEDLINE**

1. intervention.mp
2. treatment.mp
3. rehabilitation.mp
4. Cognitive therapy.mp
5. Cognitive training.mp
6. Memory training.mp
7. Computerized.mp
8. exercise.mp
9. Cognitive behavioral.mp
10. Metacognit*.mp
11. Memory self-efficacy.mp
12. Metamemory.mp
13. Goal management.mp
14. Expectancy change.mp
15. Psychological.mp
16. Group.mp
17. Cognitive.mp
18. Mindfulness.mp
19. Mediterranean diet.mp
20. Acupuncture.mp
21. Mediation.mp
22. Lifestyle.mp
23. Diet.mp
24. Gingko.mp
25. Complementary.mp
26. Vitam*.mp
27. supplement*.mp
28. acetylcholinesterase inhibitor.mp
29. Donepezil.mp
30. Memantine.mp
31. Galantamine.mp
32. Rivastigmine.mp
33. or/1-32
34. Subjective cognitive decline.mp
35. Subjective memory complaint.mp
36. Subjective cognitive concern.mp
37. Subjective cognitive complain*.mp
38. SCD.mp
39. SMC.mp
40. SCC.mp
41. preclinical dementia.mp
42. Preclinical alzheimer.mp
43. or/34-42
44. Randomized controlled trial.mp
45. Controlled trial.mp
46. Intervention trial.mp
47. Intervention study.mp
48. Experimental study.mp
49. Crossover trial.mp
50. RCT.mp
51. Parallel.mp
52. or/44-51
53. 33 AND 43 AND 52

Table 4: Interventions for people with subjective cognitive decline, search string in **CENTRAL**

#1	(“intervention”): ti,ab,kw
#2	(“treatment”): ti,ab,kw
#3	(“rehabilitation”): ti,ab,kw
#4	(“cognitive therapy”): ti,ab,kw
#5	(“cognitive training”): ti,ab,kw
#6	(“memory training”): ti,ab,kw
#7	(“computerized”): ti,ab,kw
#8	(“exercise”): ti,ab,kw
#9	(“cognitive behavioral”): ti,ab,kw
#10	(“metacognit*”): ti,ab,kw
#11	(“memory self-efficacy”): ti,ab,kw
#12	(“metamemory”): ti,ab,kw
#13	(“goal management”): ti,ab,kw
#14	(“expectancy change”): ti,ab,kw
#15	(“psychological”): ti,ab,kw
#16	(“group”): ti,ab,kw
#17	(“cognitive”): ti,ab,kw
#18	(“mindfulness”): ti,ab,kw
#19	(“mediterranean diet”): ti,ab,kw
#20	(“acupuncture”): ti,ab,kw
#21	(“mediation”): ti,ab,kw
#22	(“lifestyle”): ti,ab,kw
#23	(“diet”): ti,ab,kw
#24	(“gingko”): ti,ab,kw
#25	(“complementary”): ti,ab,kw
#26	(“vitam*”): ti,ab,kw
#27	(“supplement*”): ti,ab,kw
#28	(“acetylcholinesterase inhibitor”): ti,ab,kw
#29	(“donepezil”): ti,ab,kw
#30	(“memantine”): ti,ab,kw
#31	(“galantamine”): ti,ab,kw
#32	(“rivastigmine”): ti,ab,kw
#33	{or #1- #32}
#34	(“subjective cognitive decline”): ti,ab,kw
#35	(“subjective memory complaint”): ti,ab,kw
#36	(“subjective cognitive concern”): ti,ab,kw
#37	(“subjective cognitive complain*”): ti,ab,kw
#38	(“SCD”): ti,ab,kw
#39	(“SMC”): ti,ab,kw
#40	(“SCC”): ti,ab,kw
#41	(“preclinical dementia”). ti,ab,kw
#42	(“preclinical Alzheimer”): ti,ab,kw
#43	{or#34-#42}
#44	(“randomized controlled trial”): ti,ab,kw
#45	(“controlled trial”): ti,ab,kw
#46	(“intervention trial): ti,ab,kw
#47	(“intervention study”): ti,ab,kw
#48	(“experimental study”): ti,ab,kw
#49	(“crossover trial”): ti,ab,kw
#50	(“RCT”): ti,ab,kw
#51	(“Parallel”): ti,ab,kw
#52	{or #44-#51}
#53	#33 and #43 and #52

Table 5: Interventions for people with subjective cognitive decline, search string in **Web of Science**

#1	**TOPIC:** (“intervention”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#2	**TOPIC:** (“treatment”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#3	**TOPIC:** (“rehabilitation”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#4	**TOPIC:** (“cognitive therapy”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#5	**TOPIC:** (“cognitive training”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#6	**TOPIC:** (“memory training”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#7	**TOPIC:** (“computerized”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#8	**TOPIC:** (“exercise”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#9	**TOPIC:** (“cognitive behavioral”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#10	**TOPIC:** (“metacognit*”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#11	**TOPIC:** (“memory self-efficacy”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#12	**TOPIC:** (“metamemory”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#13	**TOPIC:** (“goal management”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#14	**TOPIC:** (“expectancy change”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#15	**TOPIC:** (“psychological”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#16	**TOPIC:** (“group”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#17	**TOPIC:** (“cognitive”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#18	**TOPIC:** (“mindfulness”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#19	**TOPIC:** (“mediterranean diet”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#20	**TOPIC:** (“acupuncture”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#21	**TOPIC:** (“mediation”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#22	**TOPIC:** (“lifestyle”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#23	**TOPIC:** (“diet”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#24	**TOPIC:** (“gingko”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#25	**TOPIC:** (“complementary”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#26	**TOPIC:** (“vitam*”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#27	**TOPIC:** (“supplement*”) Indexes=SCI-EXPANDED, SSCI Timespan=All years
#28	**TOPIC:** (“acetylcholinesterase inhibitor”) Indexes=SCI-EXPANDED, SSCI Timespan=All years

#29 **TOPIC:** (“donepezil”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#30 **TOPIC:** (“memantine”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#31 **TOPIC:** (“galantamine”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#32 **TOPIC:** (“rivastigmine”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#33 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR#10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#34 **TOPIC:** (“subjective cognitive decline”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#35 **TOPIC:** (“subjective memory complaint”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#36 **TOPIC:** (“subjective cognitive concern”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#37 **TOPIC:** (“subjective cognitive complain*”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#38 **TOPIC:** (“SCD”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#39 **TOPIC:** (“SMC”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#40 **TOPIC:** (“SCC”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#41 **TOPIC:** (“preclinical dementia”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#42 **TOPIC:** (“preclinical Alzheimer”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#43 #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41 OR #42
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#44 **TOPIC:** (“randomized controlled trial”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#45 **TOPIC:** (“controlled trial”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#46 **TOPIC:** (“intervention trial)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#47 **TOPIC:** (“intervention study”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#48 **TOPIC:** (“experimental study”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#49 **TOPIC:** (“crossover trial”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#50 **TOPIC:** (“RCT”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#51 **TOPIC:** (“Parallel”)
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#52 #44 OR #45 OR #46 OR #47 OR #48 OR #49 OR #50 OR #51
Indexes=SCI-EXPANDED, SSCI Timespan=All years
#53 #33 AND #43 AND #52
Refined by: DOCUMENT TYPES: (ARTICLE)
Indexes=SCI-EXPANDED, SSCI Timespan=All years

Table 6: Interventions for people with subjective cognitive decline, search string in **PsycInfo**

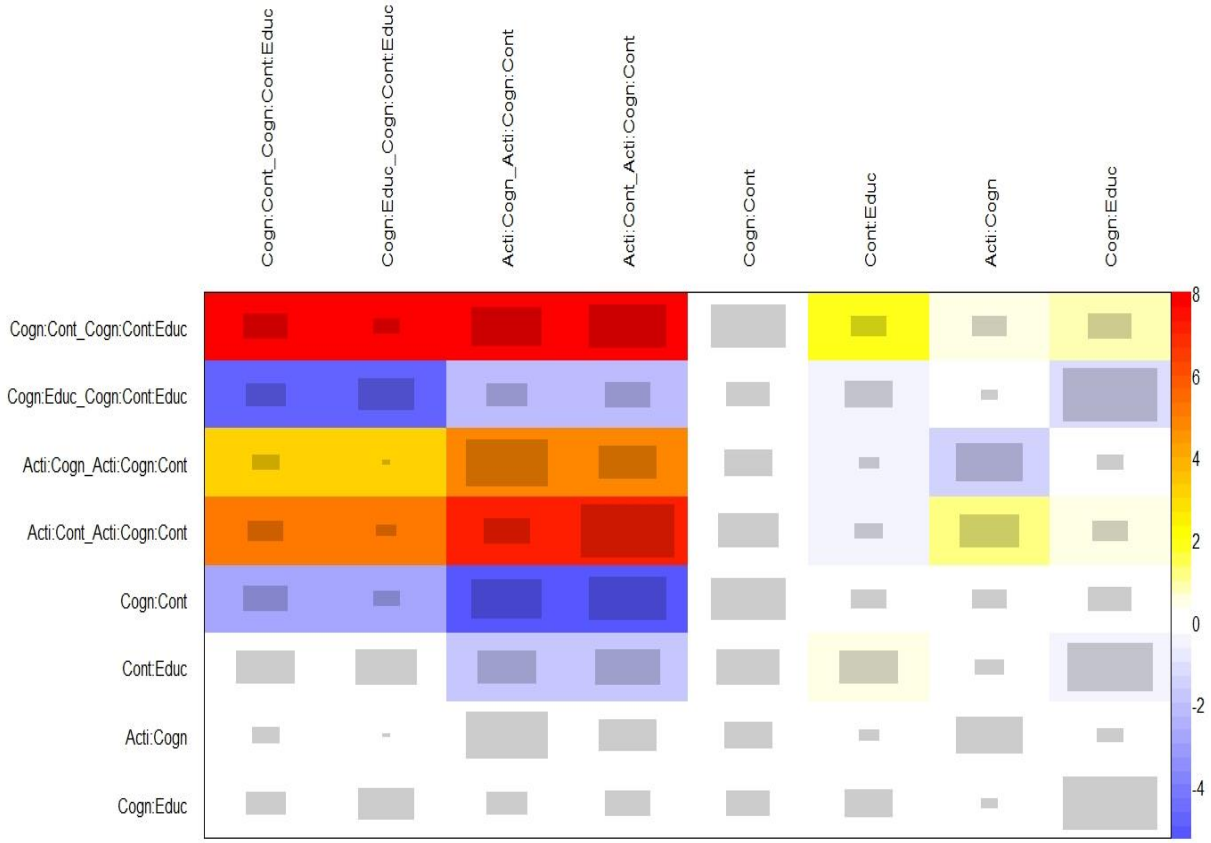
1. intervention.mp
2. treatment.mp
3. rehabilitation.mp
4. Cognitive therapy.mp
5. Cognitive training.mp
6. Memory training.mp
7. Computerized.mp
8. exercise.mp
9. Cognitive behavioral.mp
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14. Expectancy change.mp
15. Psychological.mp
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17. Cognitive.mp
18. Mindfulness.mp
19. Mediterranean diet.mp
20. Acupuncture.mp
21. Mediation.mp
22. Lifestyle.mp
23. Diet.mp
24. Gingko.mp
25. Complementary.mp
26. Vitam*.mp
27. supplement*.mp
28. acetylcholinesterase inhibitor.mp
29. Donepezil.mp
30. Memantine.mp
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48. Experimental study.mp
49. Crossover trial.mp
50. RCT.mp
51. Parallel.mp
52. or/44-51
53. 33 AND 43 AND 52

Table 7: Node-making process, definition of the clustered interventions

Node-name	**Definition**
Control Group	A passive control condition, in which participants receive a maximum of one information session or do not receive any treatment at all.
Active Control Group	An active control training that has an equal content as in the applied interventions (in these studies: cognitive training and education training), but the content of the intervention did not target the investigated outcomes. Active control training was conducted either in groups or individually.
Placebo	Placebo capsules that were orally ingested over a period of time.
Education Program	A program including several healthy life style strategies such as learning memory strategies, physical conditioning, relaxation techniques and/or diet tips. Offered either as education in small groups or in books/leaflets.
Cognitive Training	A standardized set of exercises, which involves repeated practice and is designed to impact on one or more cognitive function(s) (e.g. memory, attention). Offered either in groups or individualized settings.
Cognitive Stimulation	An intervention that involves group activities and/or discussions to enhance cognitive and social functions in general, which requires less professional guidance as e.g. cognitive training.
Physical Training	A moderate-intensity physical activity that was done either in a group setting or individually (e.g. walking, aerobics, moderate-intensity physiotherapy).
Stretching	Stretching as a specific physical training activity that only involves low-intensity stretching movements.
Mind-Body Intervention	A low-intensity physical activity in accordance with a specific philosophy that was carried out either in group sessions or individually (e.g. modification of thought processing by breathing practice, calm movements).
rTMS	Repetitive transcranial magnetic stimulation that was applied during the intervention over a specific brain region of the participants.
Cognitive Training + polyunsaturated fatty acids	A standardized set of exercises, which involved repeated practice and was designed to reflect on one or more cognitive function(s) like memory or attention. Offered either in groups or in individual trainings. Additionally, polyunsaturated fatty acids were given.
Polyunsaturated fatty acids	Supplementation with omega-3 fatty acid over a period of time.
Whey peptide	Whey peptide tablets were ingested over a period of time.
MHBA	Matured hop bitter acids capsules that were orally ingested over a period of time.
Liraglutide	Liraglutide, an insulin-sensitizing agent with neuroprotective properties, was ingested over a period of time.
Testosterone	A testosterone cream was applied in several visits in the clinic.
Spermidine	Spermidine capsules that were orally ingested over a period of time.
Brain Power Capsules	Brain Power capsules that were orally ingested over a period of time.

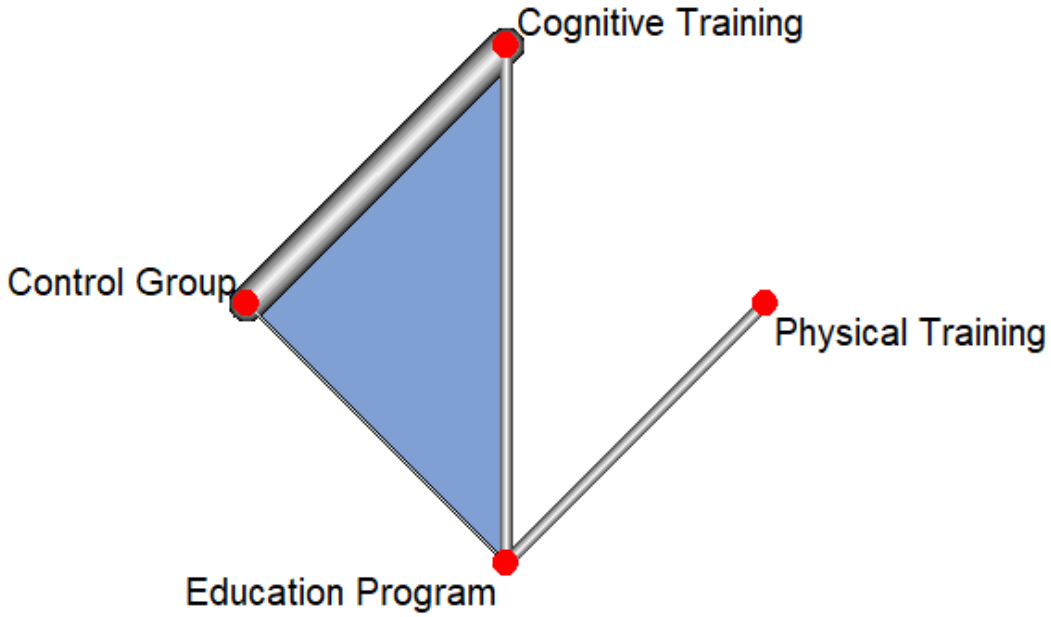
Abbreviations: rTMS = repetitive transcranial magnetic stimulation. MHBA = matured hop bitter acids.

Figure 1: Assessment of inconsistencies between direct and indirect comparisons, outcome: memory.



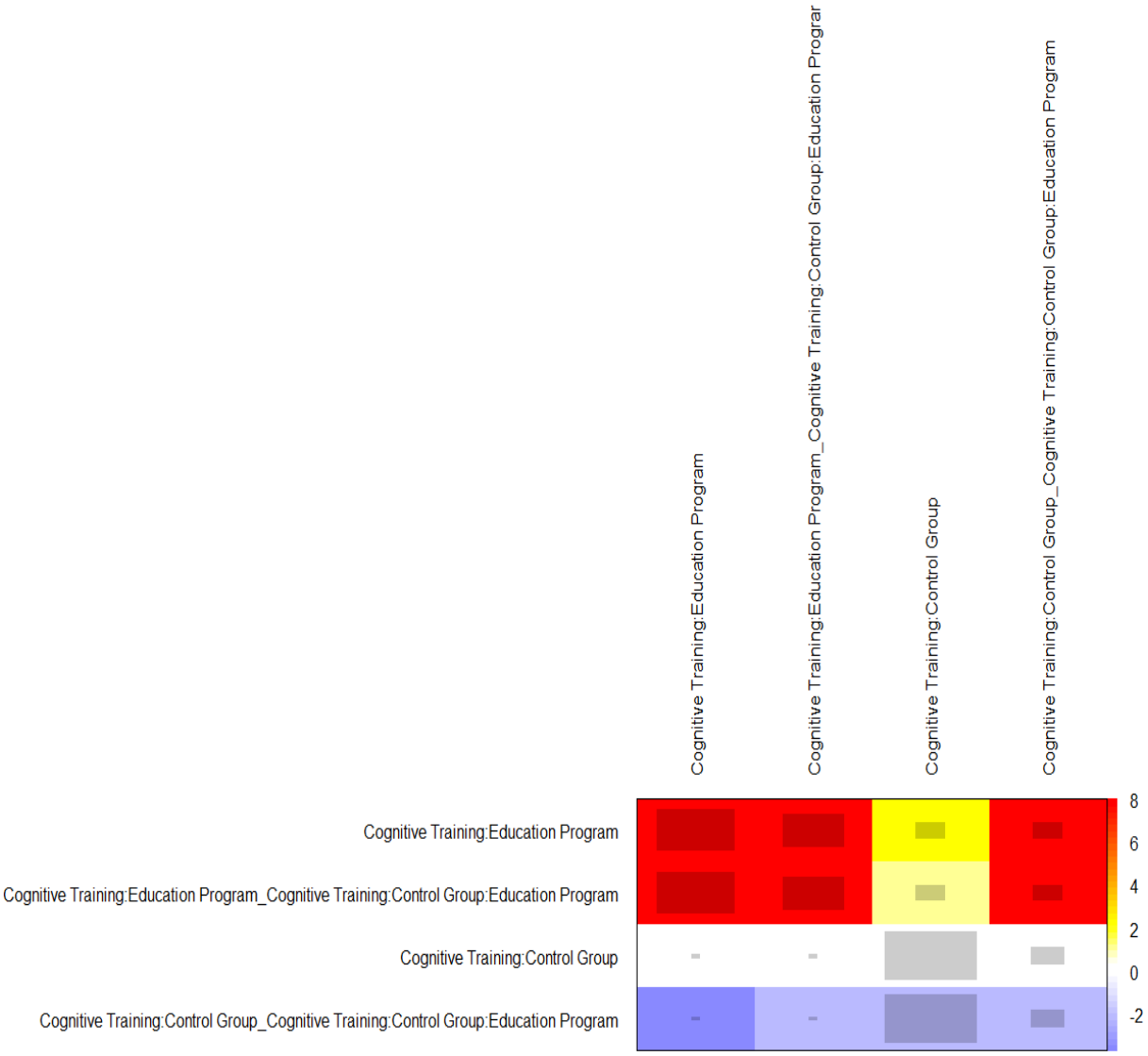
Netheat plot for assessing inconsistencies between direct and indirect evidence. The size of grey boxes indicates the importance of the treatment comparisons for the estimation of another treatment comparison. The colors, ranging from blue to red, signify the inconsistency of comparisons in a row attributable to the design in a column. Inconsistencies are displayed in the upper-left corner in red.

Figure 2. Network of analyzed comparisons for the outcome global cognition.



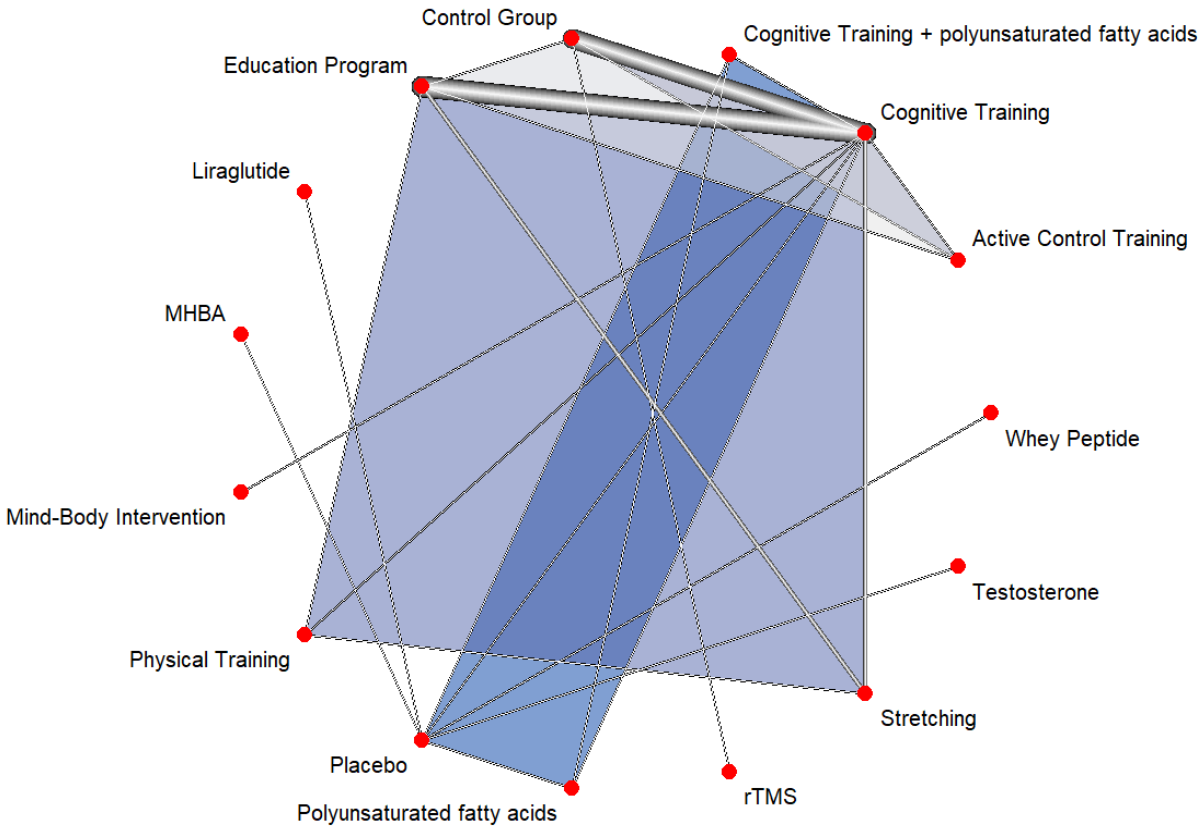
Each circle corresponds to a regimen included in the analysis. Each line represents direct comparisons between these regimens, with the thickness of the line corresponding to the number of available direct within-trial comparisons. Regimens are described in Table 7 of the Supplementary Material. Blue shaded regions correspond to studies with multiple comparisons.

Figure 3: Assessment of inconsistencies between direct and indirect comparisons, outcome: global cognition.



Netheat plot for assessing inconsistencies between direct and indirect evidence. The size of grey boxes indicates the importance of the treatment comparisons for the estimation of another treatment comparison. The colors, ranging from blue to red, signify the inconsistency of comparisons in a row attributable to the design in a column.

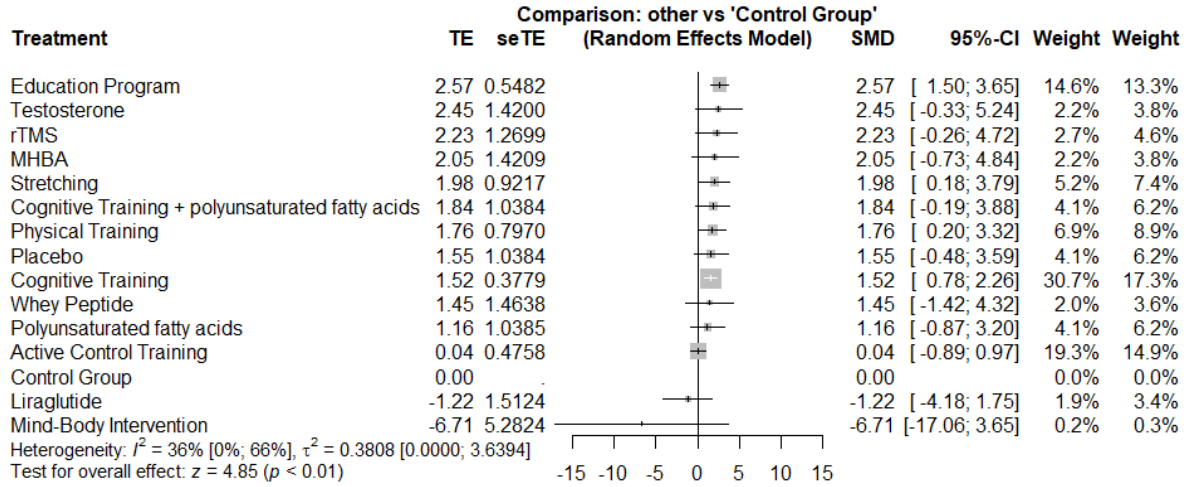
Figure 4. Network of analyzed comparisons for the outcome memory including all studies regardless of their SCD definition (sensitivity analysis).



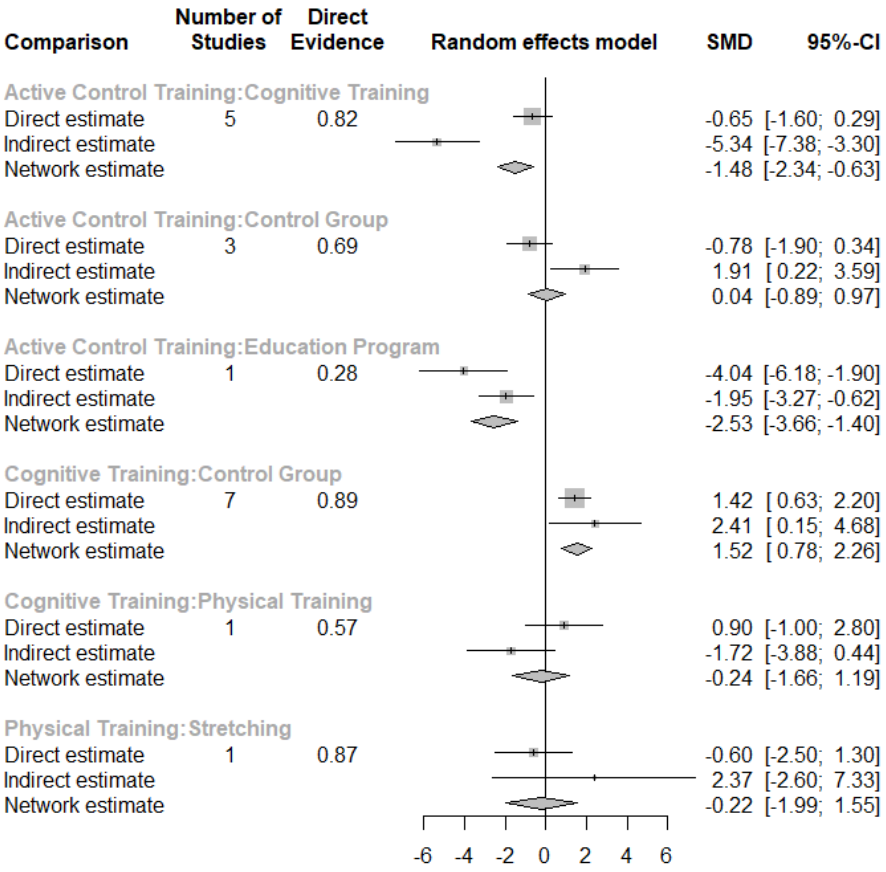
Each circle corresponds to a regimen included in the analysis. Each line represents direct comparisons between these regimens, with the thickness of the line corresponding to the number of available direct within-trial comparisons. Regimens are described in Table 7 of the Supplementary Material. Blue shaded regions correspond to studies with multiple comparisons. *Abbreviations:* rTMS = repetitive transcranial magnetic stimulation; MHBA = matured hop bitter acids.

Figure 5. Network Analyses for the outcome memory, including all studies regardless of their SCD definition (sensitivity analyses).

(A) Network analysis for the outcome memory.

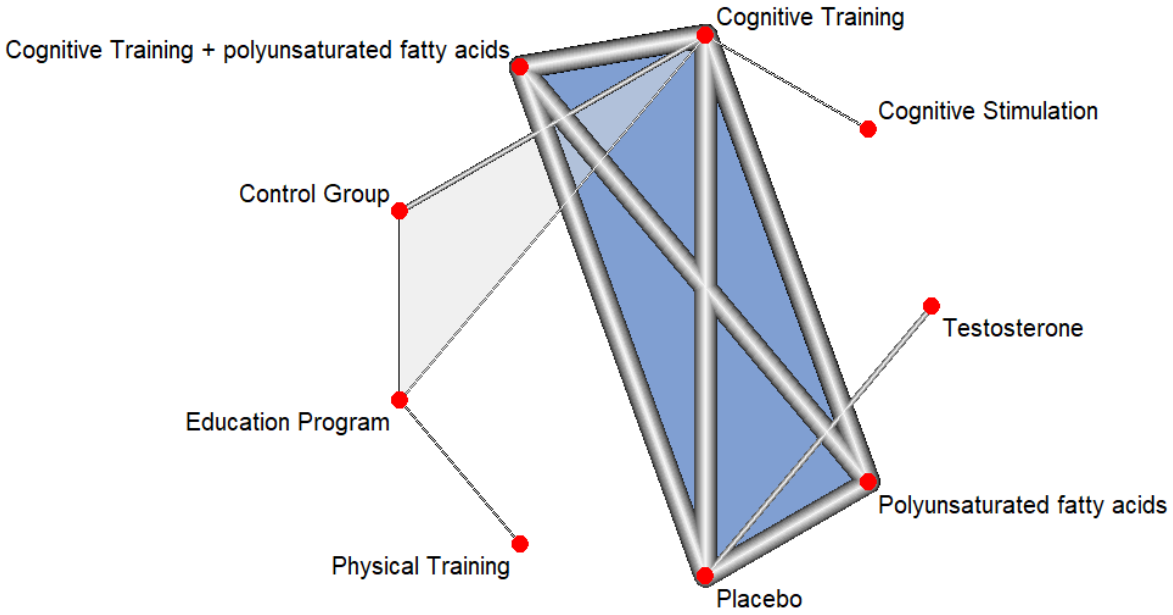


(B) Comparison of direct and indirect evidence for studies investigating memory



(A) Network Analysis of the pooled data of four treatments compared to each other and the control group reporting memory, including all studies. Treatments were ranked through P-scores. **(B)** Meta-Analysis of direct and indirect comparisons for memory. *Abbreviations:* rTMS = repetitive transcranial magnetic stimulation; MHBA = matured hop bitter acids.

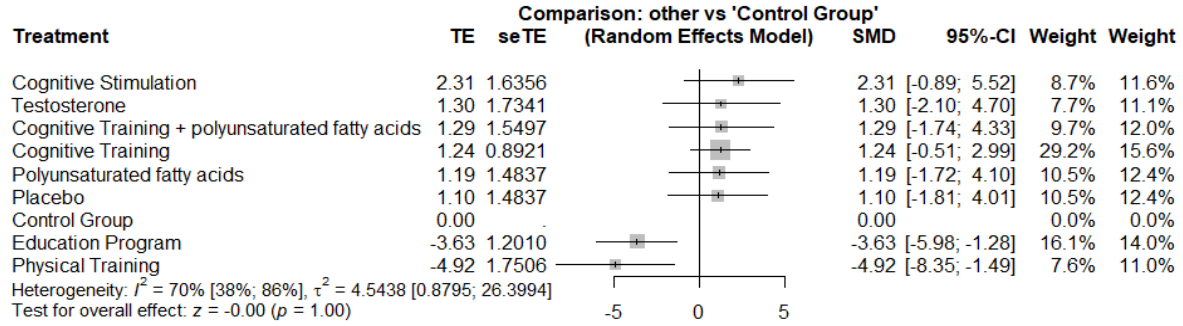
Figure 6. Network of analyzed comparisons for the outcome global cognition including all studies regardless of their SCD definition (sensitivity analysis).



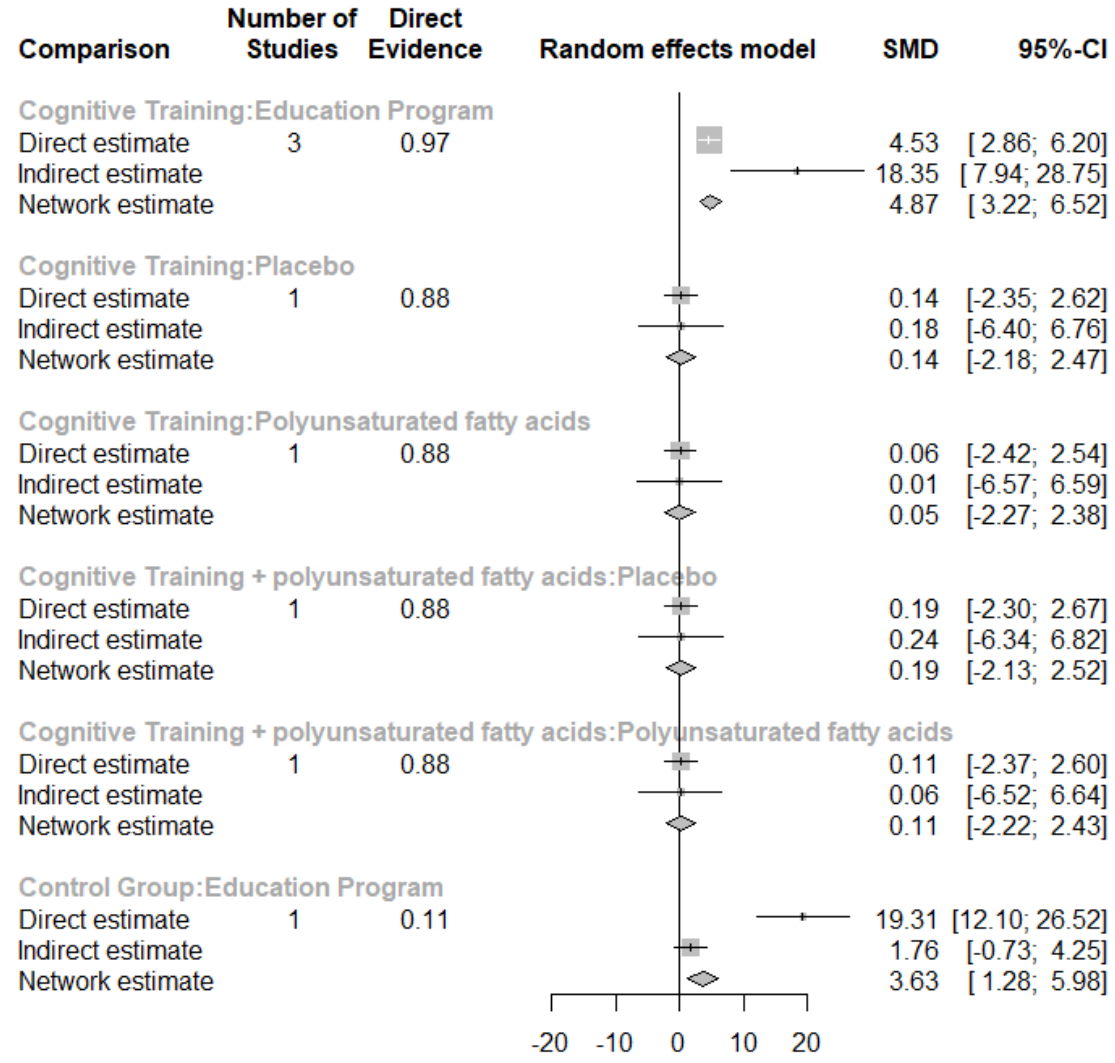
Each circle corresponds to a regimen included in the analysis. Each line represents direct comparisons between these regimens, with the thickness of the line corresponding to the number of available direct within-trial comparisons. Regimens are described in Table 7 of the Supplementary Material. Blue shaded regions correspond to studies with multiple comparisons.

Figure 7. Network Analyses for the outcome global cognition, including all studies regardless of their SCD definition (sensitivity analyses).

(A) Network analysis of the outcome global cognition.

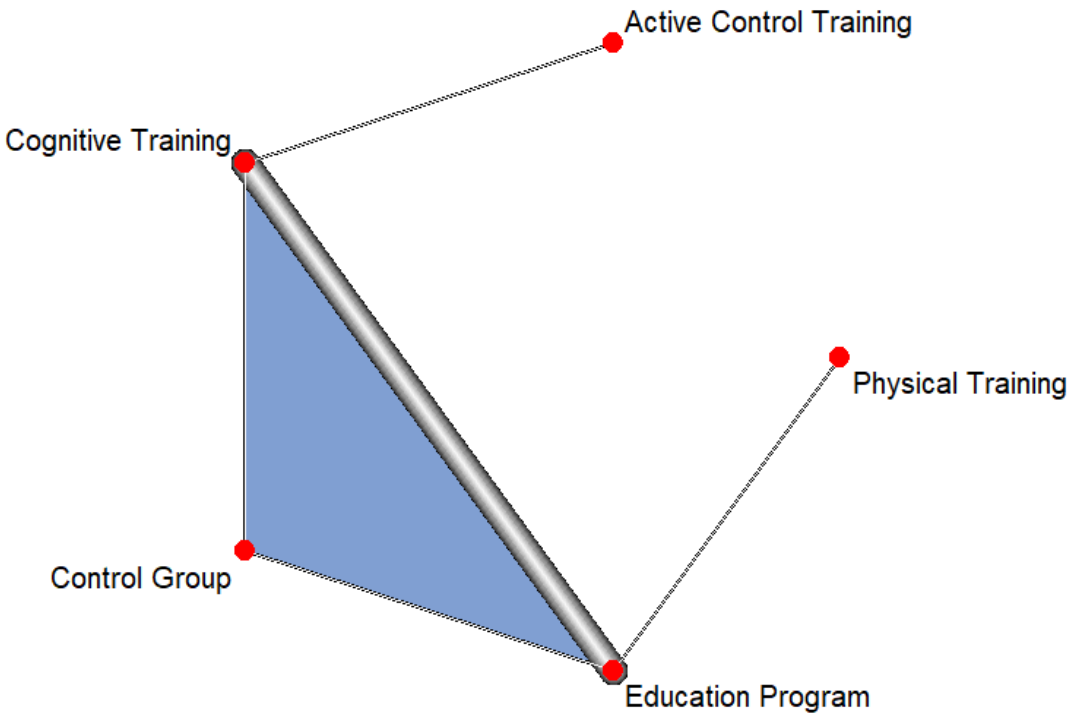


(B) Comparison of direct and indirect evidence for studies investigating global cognition.



(A) Network Analysis of the pooled data of four treatments compared to each other and the control group reporting global cognition, including all studies. Treatments were ranked through P-scores. **(B)** Meta-Analysis of direct and indirect comparisons for global cognition.

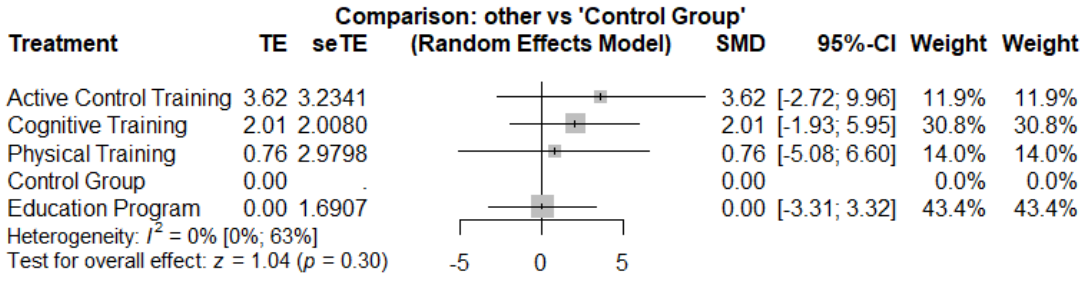
Figure 8. Network of analyzed comparisons for the outcome quality of life including all studies regardless of their SCD definition (sensitivity analysis).



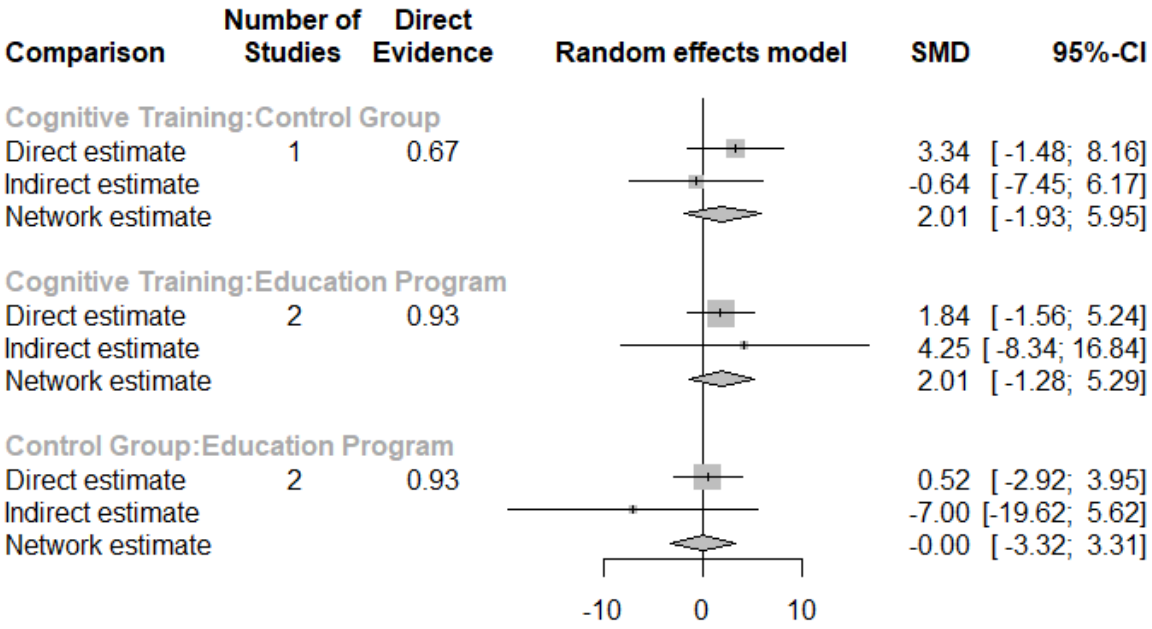
Each circle corresponds to a regimen included in the analysis. Each line represents direct comparisons between these regimens, with the thickness of the line corresponding to the number of available direct within-trial comparisons. Regimens are described in Table 7 of the Supplementary Material. Blue shaded regions correspond to studies with multiple comparisons.

Figure 9. Network Analyses for the outcome quality of life, including all studies regardless of their SCD definition (sensitivity analyses).

(A) Network analysis of the outcome quality of life.



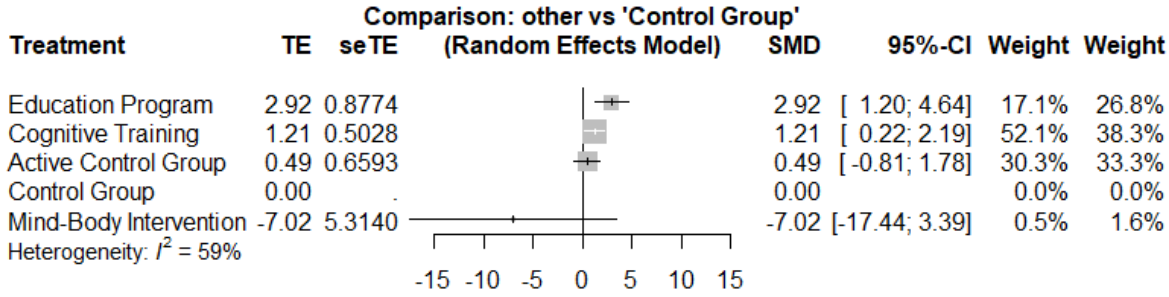
(B) Comparison of direct and indirect evidence for studies investigating quality of life.



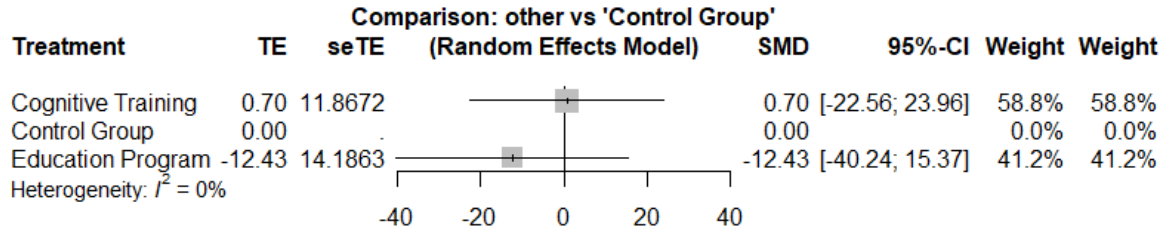
(A) Network Analysis of the pooled data of four treatments compared to each other and the control group reporting quality of life, including all studies. Treatments were ranked through P-scores. **(B)** Meta-Analysis of direct and indirect comparisons for quality of life.

Figure 10. Network Analyses for all outcomes, including studies with suitable SCD definition and low or medium risk of bias (sensitivity analyses).

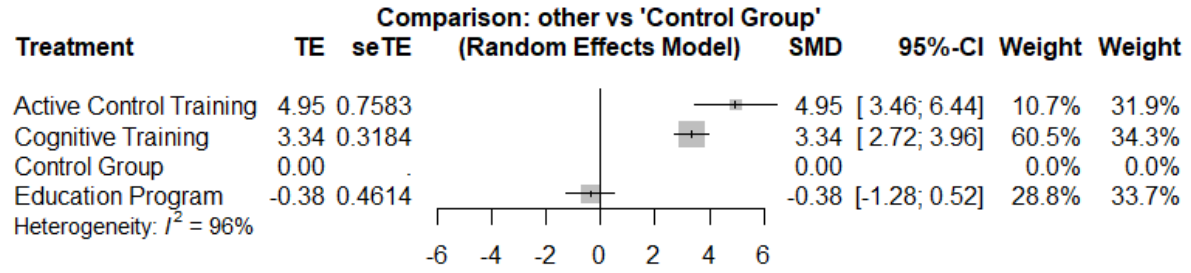
(A) Memory



(B) Global Cognition



(C) Quality of Life



(A) Network Analysis of the pooled data of five treatments compared to each other and the control group reporting **memory**, including only studies with a low to medium risk of bias and adequate SCD definition. Treatments were ranked through P-scores. **(B)** Network Analysis of the pooled data of three treatments compared to each other and the control group reporting **global cognition**, including only studies with a low to medium risk of bias and adequate SCD definition. Treatments were ranked through P-scores. **(C)** Network Analysis of the pooled data of four treatments compared to each other and the control group reporting **quality of life**, including only studies with a low to medium risk of bias and adequate SCD definition. Treatments were ranked through P-scores.

Appendix A: Detailed description of network meta-analysis calculation.

Studies with multiple treatment groups

For studies with multiple treatment groups, we combined arms as long as they could be regarded as subtypes of the same intervention [1]. When arms could not be pooled this way, we included multi-arm trials using a network meta-analysis approach that accounts for the within-study correlation between the effect sizes by re-weighting all comparisons of each multi-arm study [2–4]. The node-making process is described in Table 7 of the Supplementary Materials and was reached based on an expert consensus by the reviewers [5].

Missing Data

The following steps were implemented to address missing data (Higgins et al. 2019, Ch. 16): Whenever possible, we contacted the original investigators to request relevant missing data. If the number of participants included for a given outcome was not reported, the number of participants randomized per treatment arm was used as the denominator. If only percentages but no absolute number of events were reported for binary outcomes, we calculated numerators using percentages. If estimates for mean and standard deviations were missing, we calculated these statistics from reported data whenever possible, using approaches described in Chapter 7.7 of the Cochrane Handbook for Systematic Reviews of Interventions [1].

Assessment of heterogeneity

To evaluate the presence of clinical heterogeneity, we generated summary statistics for important clinical and methodological characteristics across all included studies (see Table 1). Within each pairwise comparison, we assessed the presence of clinical heterogeneity by visually inspecting the similarity of these characteristics.

To infer about the assumption of transitivity (i.e. the assumption that no systematic differences exist between the available comparisons other than the treatments being compared [6]), we assessed whether the included interventions are similar when they are evaluated in studies with different designs. We also compared the distribution of the potential effect modifiers across the different pairwise comparisons. For all studies, we created a table of important clinical and methodological characteristics (e.g., age, sex, and education of participants, intervention duration, SCD definition, see Table 1). We visually inspected the similarity of these factors, including the inclusion and exclusion criteria of each trial in the network. Because we aimed at including a wide range of interventions for SCD, there is substantial diversity of the investigated interventions in the network. However, because of the other narrow inclusion criteria (e.g., only studies with proper SCD definition), we expect transitivity across our treatment comparisons. We further decided to only include studies in the primary analysis that had a proper SCD definition (e.g. clear definition criteria stated in inclusion and exclusion sections) and analyzed only direct post-assessment data, because only very few studies had reported follow-up assessments.

Assessment of statistical heterogeneity and inconsistency

To evaluate the presence of statistical heterogeneity and inconsistency within the resulting networks, we used the generalized heterogeneity Q total and the generalized I^2 statistic (7). The *decomp.design* command in the R package *netmeta 1.0-1* (R Core Team 2019; Rücker 2019) was used for decomposition of the heterogeneity statistic into a Q statistic for assessing the heterogeneity between studies with the same design and a Q statistic for assessing the designs inconsistency to identify the amount of heterogeneity/inconsistency within as well as between designs. To evaluate the presence of inconsistency locally, we compared direct and indirect treatment estimates of each treatment comparisons as this can be used to inspect the consistency of a network meta-analysis [8]. For this purpose, we used the

netsplit command in the R package *netmeta 1.0-1*, which enables the splitting of the network evidence into direct and indirect contributions [9]. For each treatment comparison, direct and indirect treatment estimates and the network are presented as forest plots.

We interpreted I^2 values according to Chapter 9.5.2 of the Cochrane Handbook for Systematic Reviews of Interventions [10] as follows: 0% to 40% might not be important, 30% to 60% may represent moderate heterogeneity, 50% to 90% may represent substantial heterogeneity, 75% to 100% represents considerable heterogeneity. P-values of the Chi^2 test are only used for describing the extent of heterogeneity and not for determining statistical significance. Tau^2 was used to describe between-study variance in random-effects meta-analyses. In the event of excessive heterogeneity, unexplained by potential subgroup analyses, outcome results were not reported as pooled effect estimate of the network meta-analysis, but a narrative description of the results of each study was provided.

Assessment of reporting biases

In pairwise comparisons with at least ten trials, the presence of small-study effects was examined graphically by generating comparison adjusted funnel plots [11]. Linear regression tests [12] assessed funnel plot asymmetry. A p-value less than 0.1 was considered significant [13]. The presence of small-study effects was only assessed for the primary outcome memory. We also searched study registries, to identify completed, but not published trials.

Data synthesis: Methods for direct treatment comparisons

As pairwise comparisons are part of the network meta-analysis, additional pairwise meta-analysis were not performed. We calculated random-effects models using the R package *meta* [14] for statistical analyses.

Data synthesis: Methods for indirect and mixed comparisons

As the data was considered sufficiently similar to be combined and met the criteria of transitivity, we performed a network meta-analysis using the frequentist weighted least squared approach described by Rücker (2012). We used a random-effects model, taking into account the correlated treatment effects in multi-arm studies. We assumed common estimate for the heterogeneity variance across the different comparisons. To evaluate the extent to which treatments were connected, a network plot was provided for primary and secondary outcomes. For each comparison, the estimated treatment effect along with its 95% CI is provided. We graphically present the results using forest plots, with either Control Group or Placebo Group as reference treatment. We used the R package *netmeta 1.0-1* [9] for statistical analyses.

Sensitivity analysis

Per definition, a sensitivity analysis is a repeat of the primary analysis, substituting alternative decisions or ranges of values for decisions that were arbitrary or unclear [15]. In the present study, sensitivity analyses were conducted to test the robustness of our results by analyzing studies with low and medium risk of bias only (primary analyses; i.e., that only included studies with proper SCD definition). We judged studies as high risk, if two risk of bias domains are judged as high risk. Further, additional sensitivity analyses were conducted including all studies, regardless of their SCD definition.

Appendix B: Systematic review: Characteristics of the included studies

4,692 participants with SCD from n = 54 different studies were included in the systematic review, investigating all possible SCD interventions. The average sample size was 88 participants varying from n = 12 [16] to n = 799 participants [17] per study. The mean age of the participants was 68.98 and the age range was 40.16 to 77.30 years. Four studies did not report age of the participants [18–21]. Two studies [16, 22] only included female participants, one study only included male participants [20]. Three studies did not report the sex of the participants [19, 21, 23]. The remaining samples were mixed with an average of 66.20 % women. Mean education was 13.09 years and varied from 6.20 [24] to 17.47 years [25]. Twenty studies did not report education of their participants in years, while 12 studies did not report education of their participants at all. Sixteen studies included a follow-up. The mean follow-up interval was 6.08 months ranging from 1 week [22] to 18 months [26]. Studies included interventions using educational programs, memory techniques, cognitive training, meditation, physical exercise, nutritional supplements, pharmaceutical interventions, and non-invasive brain stimulation. In these, the following outcomes were considered: n = 20 global cognition, n = 42 memory, n = 8 quality of life, n = 7 adverse events, and n = 25 studies included other outcome measurements. An in depth overview can be found in Table 1.

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Appendix C: References of studies included in the Review

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